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Program : Computer Engineering Program Date : **22 February 2026**
Examiners : Dr. Tarek salem Duration **2 hours**
Eng :Eman,Habiba, Basant
Course/Code : Database Management Systems / CSE112

Part 1: Basic Definitions**1) What is Data?**

Data: Raw facts such as numbers, text, images, etc.
Example: Student name, ID, grade.

2) What is a Database?

A Database is a collection of related data.
It represents a small part of the real world called a Mini-world.
Example: University system (students, courses, instructors).

3) What is a DBMS?

DBMS (Database Management System) is software that helps us create, store, retrieve, update, and manage data.
Examples: MySQL, Oracle, SQL Server, PostgreSQL.

4)What is the difference between data and a database?

Data: Known facts that can be recorded and have an implicit meaning.

Database: A highly organized, interrelated, and structured set of data about a particular enterprise, controlled by a Database Management System (DBMS).

Part 2: Problems of File Systems

- Before DBMS, companies used file processing systems.
- Data Redundancy: Same data stored multiple times.
- Data Inconsistency: Data becomes incorrect if not updated everywhere.
- Data Isolation: Data stored in different formats and files.
- Difficult Maintenance: Changing file structure requires changing programs.

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Part 3: Why Do We Need Databases?

In today's world, organizations deal with large amounts of data such as student records, bank transactions, employee information, and customer details.

If data is stored in separate files without organization, several problems occur:

- **Data redundancy** (duplicate data)
- **Data inconsistency** (conflicting information)
- **Data isolation** (difficulty accessing data)
- **Poor security**
- **Difficult maintenance**

To solve these problems, we use **Databases**.

A database provides:

- Centralized and organized data storage
- Reduced redundancy
- Improved consistency
- Secure access control
- Multi-user support
- Backup and recovery mechanism

Part 4: University Database Example

- **Main Entities:**
 - STUDENT
 - COURSE
 - INSTRUCTOR
 - DEPARTMENT
 - SECTION
 - **Relationships:**
 - Students take courses
 - Instructors teach courses
 - Courses belong to departments
- Modeled using Entity Relationship (ER) Model.

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Part 5: Types of Databases

- **Traditional Applications** – Store structured numeric and textual data (e.g., student records, banking systems, inventory control). Usually implemented using relational DBMS (MySQL, Oracle, SQL Server).

- **Recent / Modern Applications** – Handle complex, large-scale, multimedia, or time-sensitive data:

Multimedia Databases (images, audio, video) → e.g., YouTube, medical imaging

Data Warehouses → historical data for analysis and decision support

Geographic Information Systems (GIS) → spatial data (maps, coordinates) → e.g., Google Maps

Real-time and Active Databases

Practical Application / Example:

Traditional: University student transcript system (Oracle).

Modern: Hospital PACS system storing X-ray images and videos (Multimedia DB).

- **Traditional Applications:**
 - Banking systems
 - Student records
 - Inventory systems
- **Modern Applications:**
 - Multimedia databases
 - Data Warehouses
 - GIS systems

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Part 6: Difference Between DB Languages and Programming Languages

❖ Programming Languages

Examples:

C++

Java

Python

Programming languages are used to:

- Build complete applications and systems
- Write algorithms and logical operations
- Control application behavior
- Implement loops, conditions, functions, and classes
- Handle user interfaces

They focus on **how** to solve a problem step by step.

Example in Java:

```
if (grade > 50) {  
    System.out.println("Pass");  
}
```

In programming languages:

You define the logic.

You control execution flow.

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❖ Database Languages (SQL)

The main database language is:

SQL (Structured Query Language)

SQL is used to:

- Create databases and tables
- Insert data
- Update data
- Delete data
- Retrieve data
- Define constraints

Example:

SELECT * FROM Students;

SQL does not focus on algorithms.

Instead, it focuses on **what data you want**, not how to get it.

This is called a **declarative language**.

When you write:

SELECT * FROM Students;

You are telling the DBMS:

"I want all students."

You are NOT telling it: How to search- How to scan memory - How to optimize

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Key Differences

Programming Languages	Database Language (SQL)
Used to build full applications	Used to manage data
Procedural / Object-Oriented	Declarative
Focus on logic & algorithms	Focus on data manipulation
Control flow (if, loop, functions)	Queries (SELECT, INSERT, UPDATE)
Example: Java, C++, Python	Example: SQL

How They Work Together

In real applications, both are used together.

For example:

Java application

Connected to MySQL database

Java sends SQL queries

Database returns results

So:

Programming language = Application logic

SQL = Data management

They complete each other.

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Part 7: Database System Environment

Database system contains:

- Users (Students, Admins)
- Application Programs
- DBMS Software
- Metadata (Data about data)
- Stored Database (Actual data)

Part 8: Setup Guide

Now we explain what we will use in the lab.

We will use:



1. Install SQL Server Management

Download:

<https://www.microsoft.com/en-us/sql-server/sql-server-downloads>

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Express Edition

Select an installation type:

Basic

Select Basic installation type to install the SQL Server Database Engine feature with default configuration.

Custom

Select Custom installation type to step through the SQL Server installation wizard and choose what you want to install. This installation type is detailed and takes longer than running the Basic install.

Download Media

Download SQL Server setup files now and install them later on a machine of your choice.

SQL Server transmits information about your installation experience, as well as other usage and performance data, to Microsoft to help improve the product. To learn more about data processing and privacy controls, and to turn off the collection of this information after installation, see the [documentation](#)

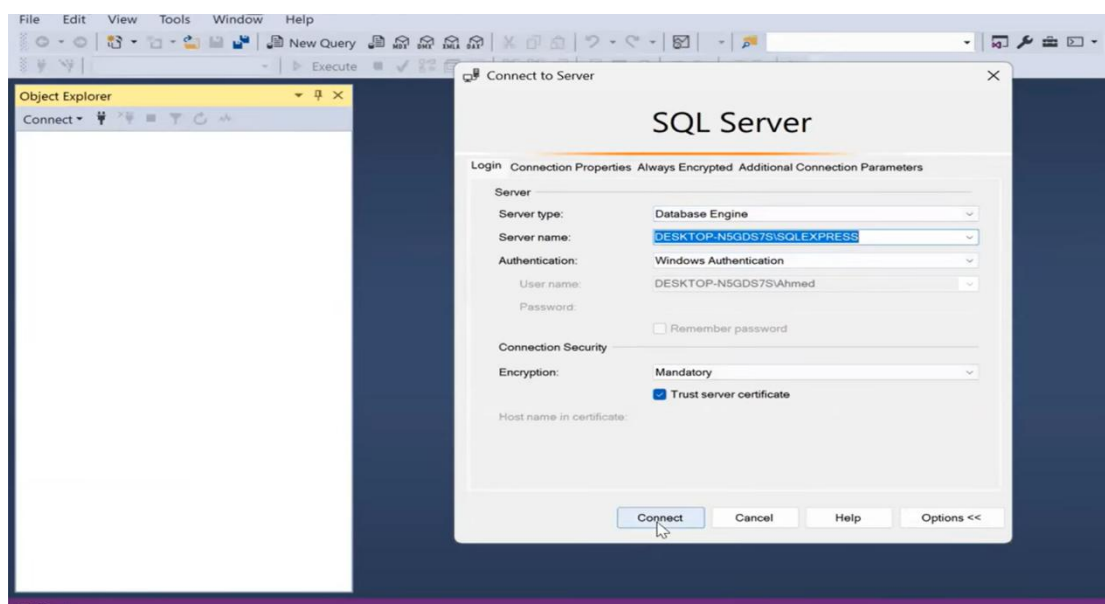
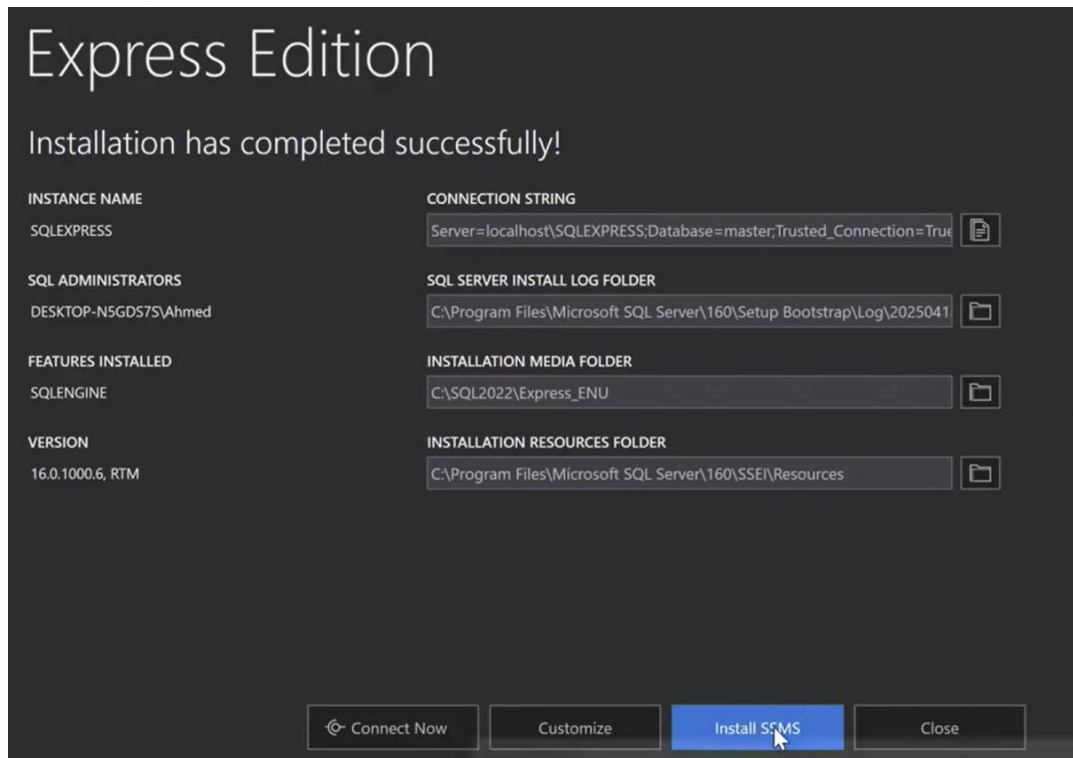
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[Install SQL Server Management Studio | Microsoft Learn](#)

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Some Questions:-

List some traditional and recent applications of databases.

Traditional applications: Numeric and textual databases.

Recent applications: Multimedia databases, Geographic Information Systems (GIS), biological and genome databases, data warehouses, mobile databases, real-time and active databases.

What are some examples of recent developments in database technology?

Social networks (e.g., Facebook, Twitter) capturing large amounts of user data.

Search engines (e.g., Google, Bing) collecting web page repositories.

Big data storage systems (e.g., distributed clusters, NOSQL systems).

Cloud-based data storage in large data centers

What is a DBMS, and what are its main functions?

DBMS (Database Management System): A software package/system to facilitate the creation and maintenance of a computerized database.

Main functions:

Define, construct, and manipulate databases.

Provide security, concurrency control, and recovery.

Support multiple user views and data abstraction

What is NOSQL, and how does it differ from traditional SQL databases?

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NOSQL (Not Only SQL): A type of database system designed to handle large volumes of unstructured or semi-structured data, often distributed across multiple servers.

Difference: NOSQL systems are more flexible in terms of data models (e.g., document-based, key-value stores) and are optimized for scalability and performance, unlike traditional SQL databases that use a rigid relational model.

What are the three levels of database system architecture?

View level: What application programs see; can hide information for security.

Logical level: Describes the structure of the entire database for users.

Physical level: Describes how data is stored on the storage medium.